ASIC Design and Automation Project

Project 2: MIPS Processor Back-End-Design

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Chapter 1 Introduction

This **project aims** to improve the back-end design skills and optimization techniques.

And output a **GDS** “layout” for MIPS16 processor from ICC VM.

The primary objectives of this project are:

-To analyze common backend architectures and their performance characteristics.

-To identify key performance bottlenecks.

-To study and apply optimization techniques such as asynchronous processing, and load balancing.

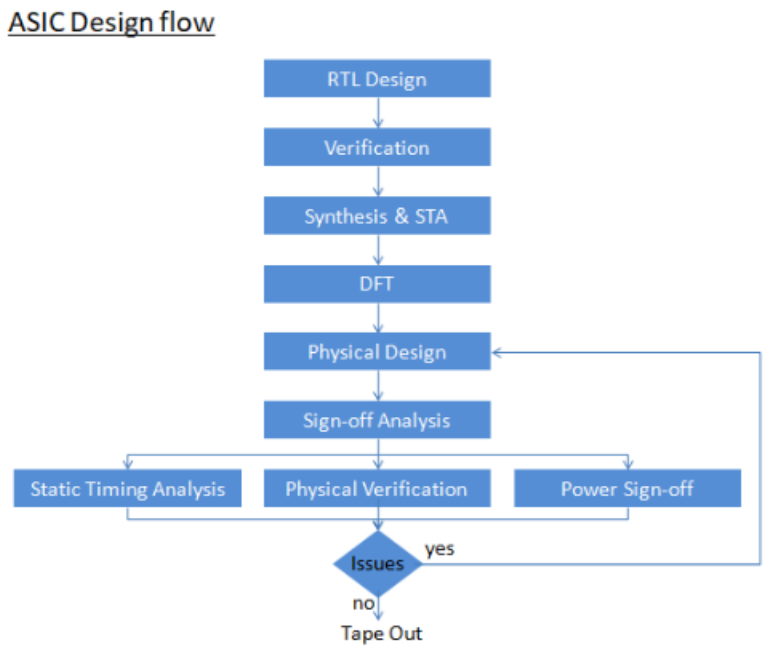
-To evaluate the impact of these techniques on system performance, scalability, and resource utilization.

-To propose best practices for designing optimized systems.

The goal is to **achieve** a design that satisfies predefined **area limits without violating timing** or functional requirements. Various optimization strategies are explored, including logic optimization, resource sharing, **constraint tuning**, and **physical design adjustments**. These techniques are evaluated throughout the backend flow to identify trade-offs between area, performance, and design complexity.

Chapter 2 Steps of Back-End Design

ASIC flow



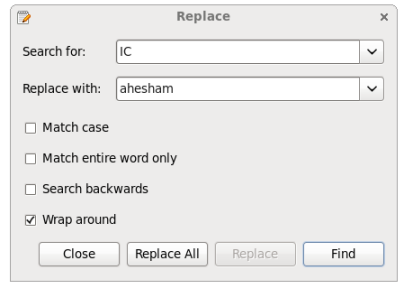
-Rtl > syn > formality > DFT > formality > floorplan > power plan > placement > cts > routing > chip finish > checks > GDS.

Synthesis

- change the date of the VM to 2018 so it can run smoothly with the license.

-1st we corrected the path in the syn.tcl to be ahesham not IC as it was to be able to see the folders and std cell libraries. So we searched for IC and changed every appearance of it with ahesham

```
set design mips_16

#set top_module mips_16_TOP

set_svf /home/ahesham/ref_flow_before/syn/output/${design}.svf

set_app_var search_path "/home/ahesham/ref_flow_before/standard_cell_libraries/NangateOpenCellLibrary_PDKv1_3_v2010_12/lib/Front_End/Liberty
set_app_var link_library "* NangateOpenCellLibrary_ss0p95vn40c.db"
set_app_var target_library "NangateOpenCellLibrary_ss0p95vn40c.db"


sh rm -rf work
sh mkdir -p work
define_design_lib work -path ./work


analyze -library work -format verilog /home/ahesham/ref_flow_before/rtl/${design}.v
elaborate $design -lib work
current_design

check_design
source /home/ahesham/ref_flow_before/syn/cons/cons.tcl
link

#compile -ungroup_all -map_effort low
compile -area_effort high -map_effort high
```



-2nd we changed the command of the effort to be high

```
#compile -ungroup_all -map_effort low
compile -area_effort high -map_effort high
```

-3rd from the file cons.tcl we changed the clock frequency to be 400MHZ as said in the requirement so the period will be 2.5ns , and make the duty cycle = 50%

```
#for 400Mhz clock we need the period to be 2.5ns
create_clock -name clk -period 2.5 -waveform {0 1.25} [get_ports clk]
```

-4th results after running the syn.tcl

Area report

```
****************************************
Report : area
Design : mips_16
Version: G-2012.06-SP2
Date   : Tue Dec 18 17:11:48 2018
****************************************

Information: Updating design information... (UID-85)
Warning: Design 'mips_16' contains 1 high-fanout nets. A fanout number of 10
Library(s) Used:

    NangateOpenCellLibrary_ss0p95vn40c (File: /home/ahesham/ref_flow_before/

Number of ports:                          34
Number of nets:                          273
Number of cells:                         181
Number of combinational cells:           158
Number of sequential cells:               16
Number of macros:                          0
Number of buf/inv:                        27
Number of references:                     22

Combinational area:       17467.688128
Buf/Inv area:             3025.750006
Noncombinational area:    19288.191353
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:          36755.879481
Total area:                 undefined
1
```

The 1 here indicates that the report is successful with no errors and if we searched the syn.log with errors you will find nothing.

Timing Report

```
------------------------------------------------------------
data required time                                      2.36
data arrival time                                      -0.79
------------------------------------------------------------
slack (MET)                                             1.57
```

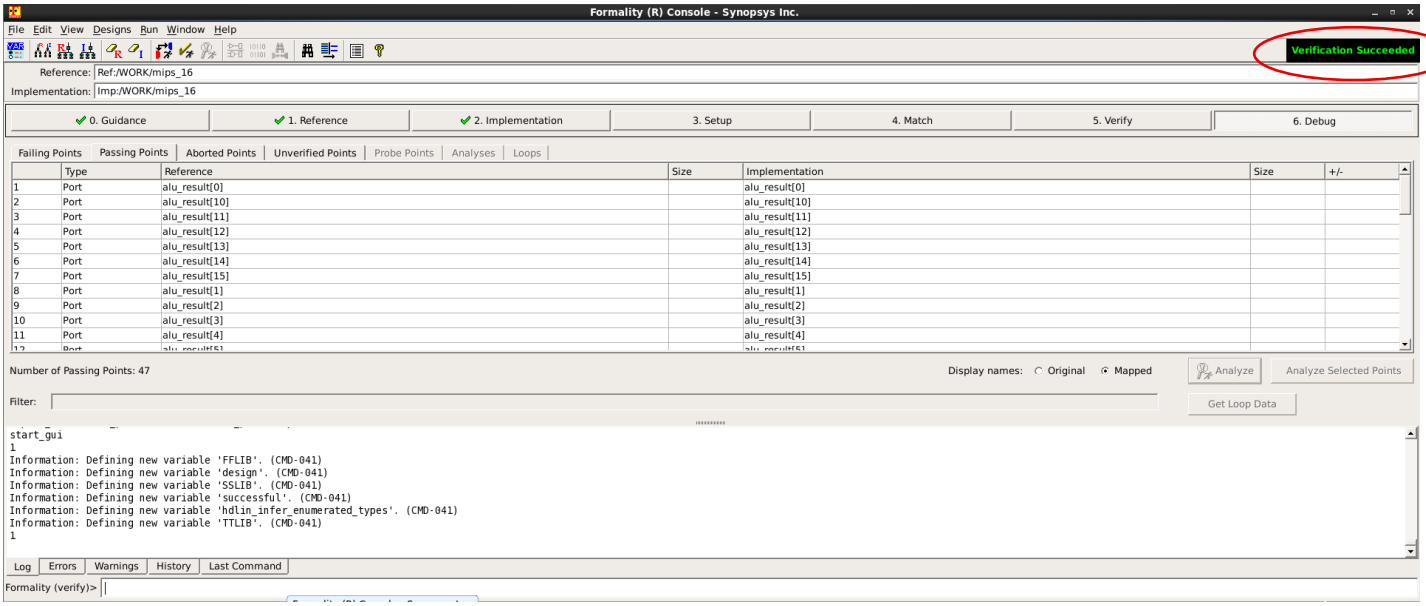
```
------------------------------------------------------------
data required time                                      2.37
data arrival time                                      -0.78
------------------------------------------------------------
slack (MET)                                             1.59
```

```
------------------------------------------------------------------------
slack (MET)                                                        1.83


1
```

```
syn.log   synth_timing.rpt

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 10
Design : mips_16
Version: G-2012.06-SP2
Date   : Tue Dec 18 17:11:49 2018
****************************************

 # A fanout number of 1000 was used for high fanout net computations.

Operating Conditions: worst_low   Library: NangateOpenCellLibrary_ss0p95vn40c
Wire Load Model Mode: top

  Startpoint: pc_current_reg[1]
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: pc_current_reg[15]
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  mips_16            5K_hvratio_1_1        NangateOpenCellLibr

  Point                                    Incr       Path
  -----------------------------------------------------------
  clock clk (rise edge)                    0.00       0.00
  clock network delay (ideal)              0.00       0.00
  pc_current_reg[1]/CK (DFFR_X1)           0.00 #     0.00 r
  pc_current_reg[1]/Q (DFFR_X1)            0.10       0.10 r
  U294/ZN (AND2_X1)                        0.05       0.15 r
  U295/ZN (AND2_X1)                        0.04       0.19 r
  U305/ZN (AND2_X1)                        0.04       0.23 r
  U296/ZN (AND2_X1)                        0.04       0.28 r
  U306/ZN (AND2_X1)                        0.04       0.32 r
  U297/ZN (AND2_X1)                        0.04       0.37 r
  U298/ZN (AND2_X1)                        0.04       0.41 r

[Terminal]   log - File Browser   synth_timing.rpt (~/ref...

Find
Search for: violations
Match case
Match entire word only
Search backwards
Wrap around
Close   Find
```

The timing is run on worst condition to see whether there is any violations or not. There is no violations since the slack is +ve.

-5th run formality



	Type	Reference	Size	Implementation	Size	+/-
1	Port	alu_result[0]		alu_result[0]		
2	Port	alu_result[10]		alu_result[10]		
3	Port	alu_result[11]		alu_result[11]		
4	Port	alu_result[12]		alu_result[12]		
5	Port	alu_result[13]		alu_result[13]		
6	Port	alu_result[14]		alu_result[14]		
7	Port	alu_result[15]		alu_result[15]		
8	Port	alu_result[1]		alu_result[1]		
9	Port	alu_result[2]		alu_result[2]		
10	Port	alu_result[3]		alu_result[3]		
11	Port	alu_result[4]		alu_result[4]		

We verified that the functionality after synthesis did not change.

```
****************************************************
Report          : unverified_points

Reference       : Ref:/WORK/mips_16
Implementation  : Imp:/WORK/mips_16
Version         : G-2012.06-SP2
Date            : Mon Dec 18 23:48:54 2017
****************************************************

No unverified compare points.

1
```

-6th pnr steps

The **area before updating** the utilization

```
CELL INSTANCE SECTION
---------------------

 Cell Instance Type        Count          Area
  Placed Cells             96332      146543.12
    Fixed Cells             3575        1901.90
    Soft Fixed Cells          40          95.76
  Unplaced Cells               0           0.00

UTILIZATION RATIOS
------------------
Chip area           : 166145.54
Core area           : 146543.12
  SiteRow area      : 146543.12
Cell/Core Ratio     : 98.7022%
Cell/Chip Ratio     : 87.0569%
(A) Logical cell sites (non-fixed) :       150681
(B) Logical cell sites (fixed) :         2383
(C) All blocked sites :          9533
(D) Total core sites :        550914
Cell Utilization(non-fixed) = 27.83% (A) / (D - C)
                                        (150681) / (550914 - 9533)
Cell Utilization(non-fixed + fixed) = 28.15% (A + B) / (D - (C - B))
                                        (150681 + 2383) / (550914 - (9533 - 2383))
Blockage Percentage = 1.73% (C) / (D)
                                        (9533) / (550914)
```

So we edited the core utilization to meet the specs(area) from 0.25 to 0.45

```
## Create Starting Floorplan
###########################
create_floorplan -core_utilization 0.45 \
        -start_first_row -flip_first_row \
        -left_io2core 12.4 -bottom_io2core 12.4 -right_io2core 12.4 -top_io2core 12.4
```

Area after editing utilization

```
CELL INSTANCE SECTION
---------------------

 Cell Instance Type        Count           Area
  Placed Cells             73505       81613.06
    Fixed Cells             2061        1096.45
    Soft Fixed Cells          40          95.76
  Unplaced Cells               0           0.00

UTILIZATION RATIOS
------------------
Chip area            : 96397.82
Core area            : 81613.06
  SiteRow area       : 81613.06
Cell/Core Ratio      : 98.6565%
Cell/Chip Ratio      : 83.5253%
(A) Logical cell sites (non-fixed) :        150504
(B) Logical cell sites (fixed) :          2347
(C) All blocked sites :          6469
(D) Total core sites :        306816
Cell Utilization(non-fixed) = 50.11% (A) / (D - C)
                                          (150504) / (306816 - 6469)
Cell Utilization(non-fixed + fixed) = 50.50% (A + B) / (D - (C - B))
                                          (150504 + 2347) / (306816 - (6469 - 2347))
Blockage Percentage = 2.11% (C) / (D)
                                          (6469) / (306816)
```

As we see the area met the specs in the report 96,397um^2 **less than 100,000um^2**

Utlilization: is the area of cells divided by the area of the block containing them. increasing utilization leads to **more crowded routing**, while decreasing it leads to a more relaxed routing (which prevents the DRCs violations) but increases the overall area.

Setup & Hold report (max path is for setup -min for Hold)

pnr.tcl | pnr.tcl | pnr_utilization.rpt | pnr_timing_setup.rpt

```
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: pc_current_reg_15_
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: max

  Point                                       Incr       Path
  -----------------------------------------------------------
  clock clk (rise edge)                       0.00       0.00
  clock network delay (propagated)            0.15       0.15
  pc_current_reg_1_/CK (DFFR_X1)              0.00       0.15 r
  pc_current_reg_1_/Q (DFFR_X1)               0.12       0.26 r
  U257/ZN (AND2_X1)                           0.05 &     0.31 r
  U258/ZN (AND2_X1)                           0.04 &     0.35 r
  U259/ZN (AND2_X1)                           0.04 &     0.39 r
  U260/ZN (AND2_X1)                           0.04 &     0.42 r
  U261/ZN (AND2_X1)                           0.04 &     0.46 r
  U262/ZN (AND2_X1)                           0.04 &     0.50 r
  U263/ZN (AND2_X1)                           0.04 &     0.54 r
  U264/ZN (AND2_X1)                           0.04 &     0.58 r
  U265/ZN (AND2_X1)                           0.04 &     0.62 r
  U266/ZN (AND2_X1)                           0.04 &     0.66 r
  U267/ZN (AND2_X1)                           0.04 &     0.69 r
  U268/ZN (AND2_X1)                           0.04 &     0.73 r
  U250/ZN (AND2_X1)                           0.04 &     0.77 r
  U245/ZN (XNOR2_X1)                          0.05 &     0.81 r
  U201/ZN (OAI221_X1)                         0.04 &     0.85 f
  pc_current_reg_15_/D (DFFR_X1)              0.00 &     0.85 f
  data arrival time                                      0.85

  clock clk (rise edge)                       2.50       2.50
  clock network delay (propagated)            0.15       2.65
  clock uncertainty                          -0.10       2.55
  pc_current_reg_15_/CK (DFFR_X1)             0.00       2.55 r
  library setup time                         -0.03       2.52
  data required time                                     2.52
  -----------------------------------------------------------
  data required time                                     2.52
  data arrival time                                     -0.85
  -----------------------------------------------------------
  slack (MET)                                            1.67
```

pnr_timing_hold.rpt (~/ref_flow_before/pnr/log) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

pnr.tcl | pnr.tcl | pnr_utilization.rpt | pnr_timing_setup.rpt | pnr_timing_hold.rpt

```
Version: G-2012.06-ICC-SP2
Date   : Sun Dec 23 10:19:35 2018
****************************************

 * Some/all delay information is back-annotated.

Operating Conditions: worst_low   Library: NangateOpenCellLibrary_ss0p95vn40c
        Parasitic source    : LPE
        Parasitic mode      : RealRC
        Extraction mode     : MIN_MAX
        Extraction derating : -40/-40/-40

Information: Percent of Arnoldi-based delays =  0.00%

  Startpoint: datamem/ram_reg_71__11_
              (rising edge-triggered flip-flop clocked by clk)
  Endpoint: datamem/ram_reg_71__11_
            (rising edge-triggered flip-flop clocked by clk)
  Path Group: clk
  Path Type: min

  Point                                       Incr       Path
  -----------------------------------------------------------
  clock clk (rise edge)                       0.00       0.00
  clock network delay (propagated)            0.19       0.19
  datamem/ram_reg_71__11_/CK (DFF_X1)         0.00       0.19 r
  datamem/ram_reg_71__11_/Q (DFF_X1)          0.09       0.28 r
  datamem/U3223/ZN (AOI22_X1)                 0.01 &     0.29 f
  datamem/U3222/ZN (INV_X1)                   0.01 &     0.30 r
  datamem/ram_reg_71__11_/D (DFF_X1)          0.00 &     0.30 r
  data arrival time                                      0.30

  clock clk (rise edge)                       0.00       0.00
  clock network delay (propagated)            0.19       0.19
  clock uncertainty                           0.10       0.29
  datamem/ram_reg_71__11_/CK (DFF_X1)         0.00       0.29 r
  library hold time                           0.02       0.30
  data required time                                     0.30
  -----------------------------------------------------------
  data required time                                     0.30
  data arrival time                                     -0.30
  -----------------------------------------------------------
  slack (MET)                                            0.00
```

There is no setup or hold violation.

DRC report

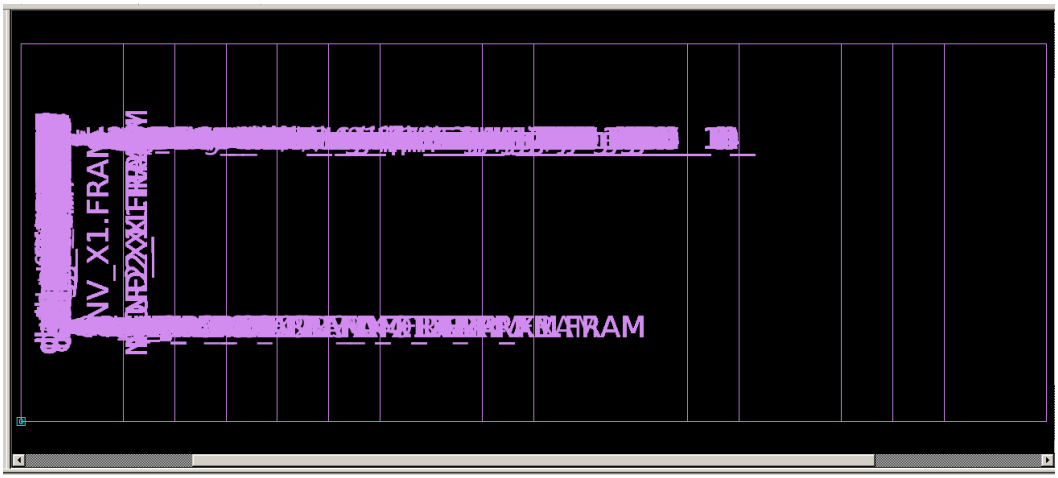
pnr.tcl | pnr.tcl | pnr_utilization.rpt | pnr_timing_setup.rpt | pnr_drc.rpt

```
DRC-SUMMARY:
       @@@@@@@ TOTAL VIOLATIONS =       0


Total Wire Length =                     369984 micron
Total Number of Contacts =              187445
Total Number of Wires =                 241506
Total Number of PtConns =               71396
Total Number of Routable Wires =        241506
Total Routable Wire Length =            365310 micron
```

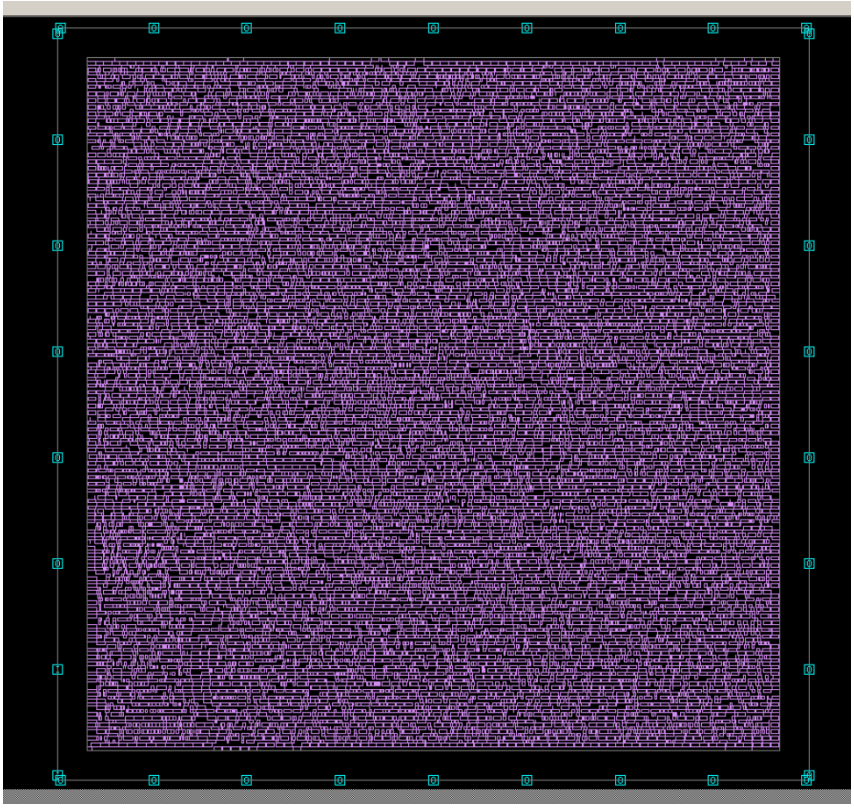
```
Verify Summary:

Total number of nets = 23121, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 0
Total number of antenna violations = no antenna rules defined
Total number of voltage-area violations = no voltage-areas defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

Screen shots of GUI

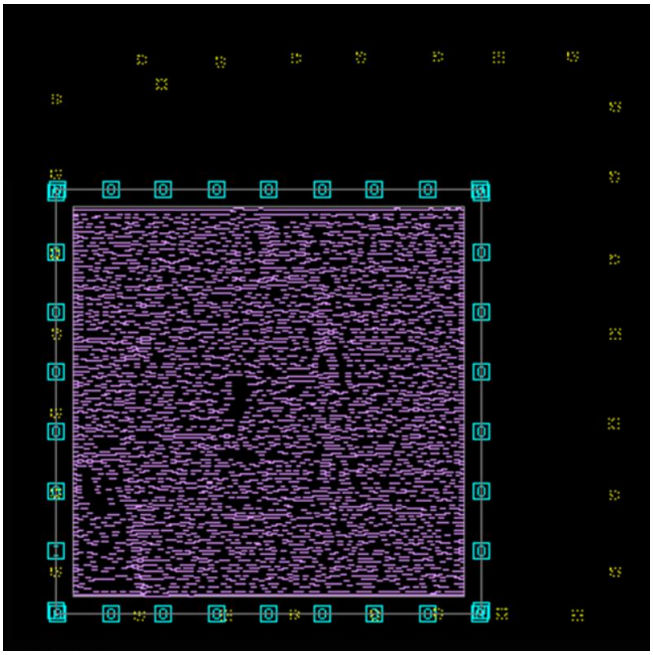
Design load



Floorplan (hiding metal layers)

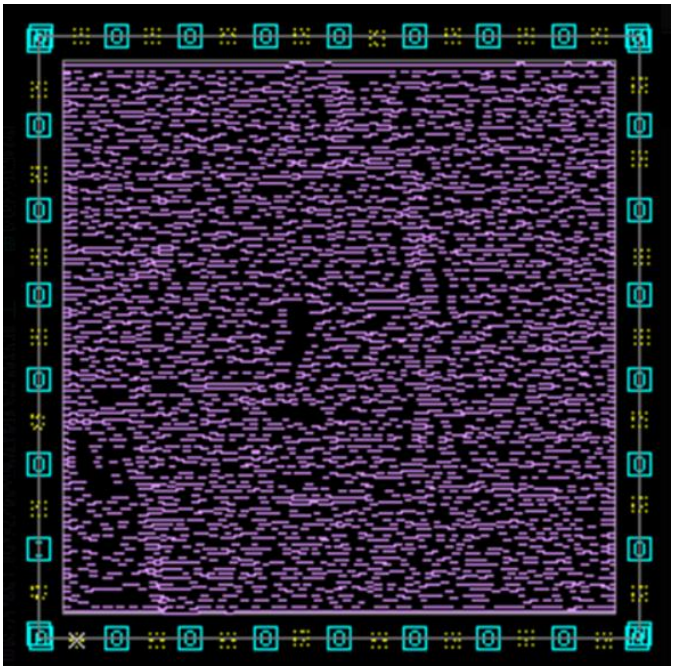


After placement of virtual pads



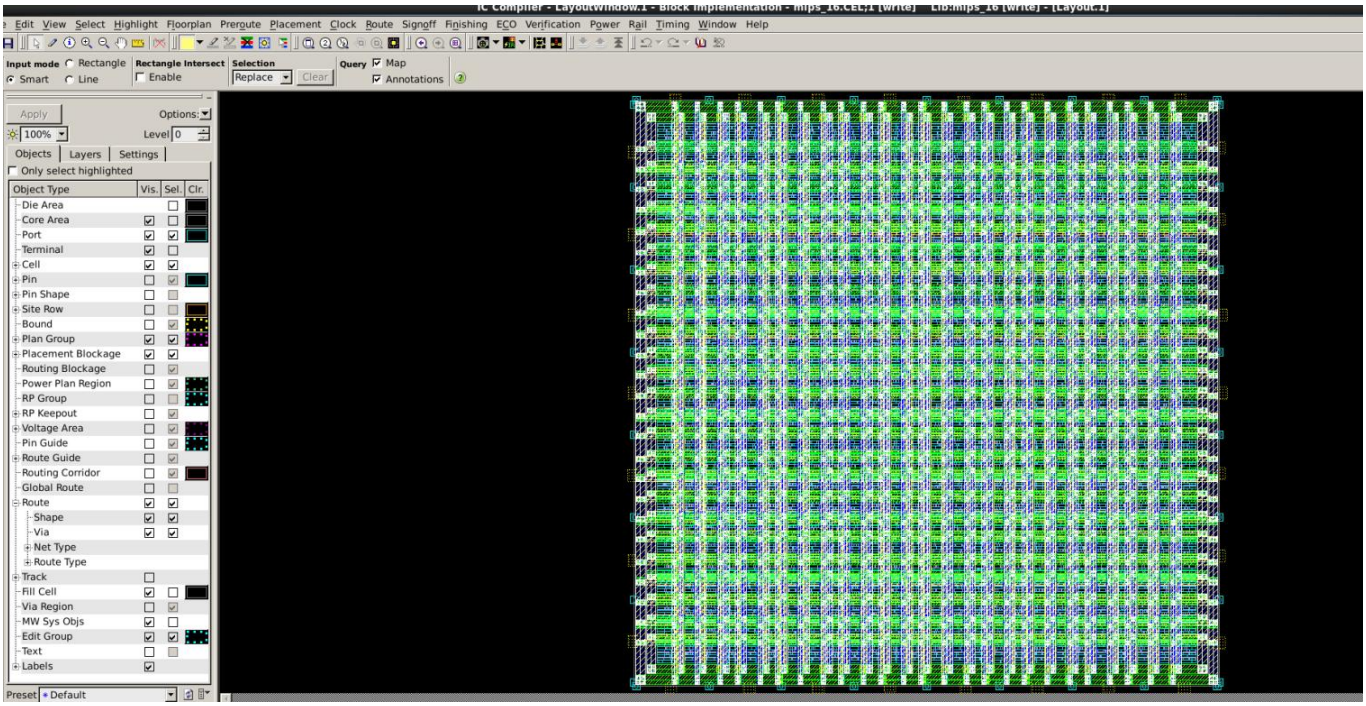
It looks like that since we increased the utilization, so the area decreased and the pads coordinates is fixed in the script

So we edited the pads location in the GUI manually and then added the commands for it to the script

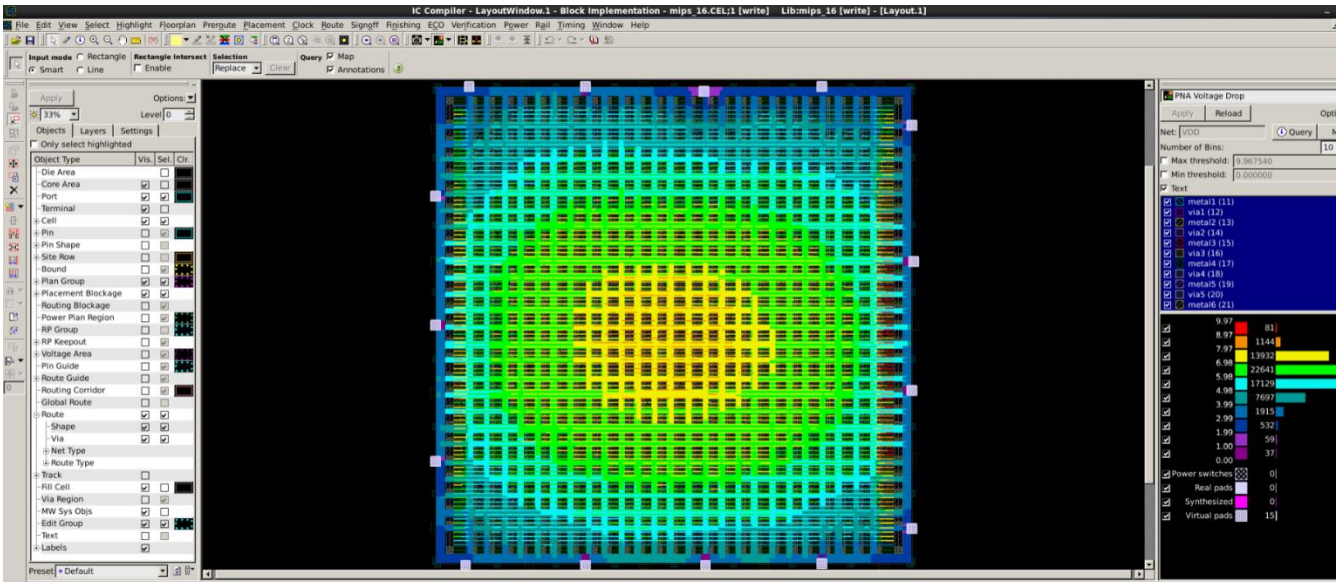


```
remove_fp_virtual_pad -all

#####################################

create_fp_virtual_pad -net VSS -point {0.0000 23.5615}
create_fp_virtual_pad -net VSS -point {-0.5745 111.5100}
create_fp_virtual_pad -net VSS -point {0.0000 196.5845}
create_fp_virtual_pad -net VSS -point {0.0000 282.8085}
create_fp_virtual_pad -net VSS -point {58.6325 310.4000}
create_fp_virtual_pad -net VSS -point {135.6590 310.4000}
create_fp_virtual_pad -net VSS -point {213.8355 310.4000}
create_fp_virtual_pad -net VSS -point {289.7125 310.4000}
create_fp_virtual_pad -net VSS -point {310.4060 247.1690}
create_fp_virtual_pad -net VSS -point {311.5555 155.1970}
create_fp_virtual_pad -net VSS -point {310.4060 68.9730}
create_fp_virtual_pad -net VSS -point {292.0115 -1.1555}
create_fp_virtual_pad -net VSS -point {213.8355 -1.1555}
create_fp_virtual_pad -net VSS -point {135.6590 -0.0060}
create_fp_virtual_pad -net VSS -point {60.9315 -1.1555}

create_fp_virtual_pad -net VDD -point {0.0000 66.6735}
create_fp_virtual_pad -net VDD -point {0.0000 155.1970}
create_fp_virtual_pad -net VDD -point {0.0000 239.1215}
create_fp_virtual_pad -net VDD -point {21.8435 310.4000}
create_fp_virtual_pad -net VDD -point {97.7205 310.4000}
create_fp_virtual_pad -net VDD -point {174.7470 309.2505}
create_fp_virtual_pad -net VDD -point {251.7740 310.4000}
create_fp_virtual_pad -net VDD -point {310.4060 285.1075}
create_fp_virtual_pad -net VDD -point {311.5555 196.5845}
create_fp_virtual_pad -net VDD -point {310.4060 112.6600}
create_fp_virtual_pad -net VDD -point {310.4060 22.9870}
create_fp_virtual_pad -net VDD -point {251.7740 -0.0060}
create_fp_virtual_pad -net VDD -point {175.8970 -1.1555}
create_fp_virtual_pad -net VDD -point {97.7205 -1.1555}
create_fp_virtual_pad -net VDD -point {19.5440 -1.1555}
```

Power network

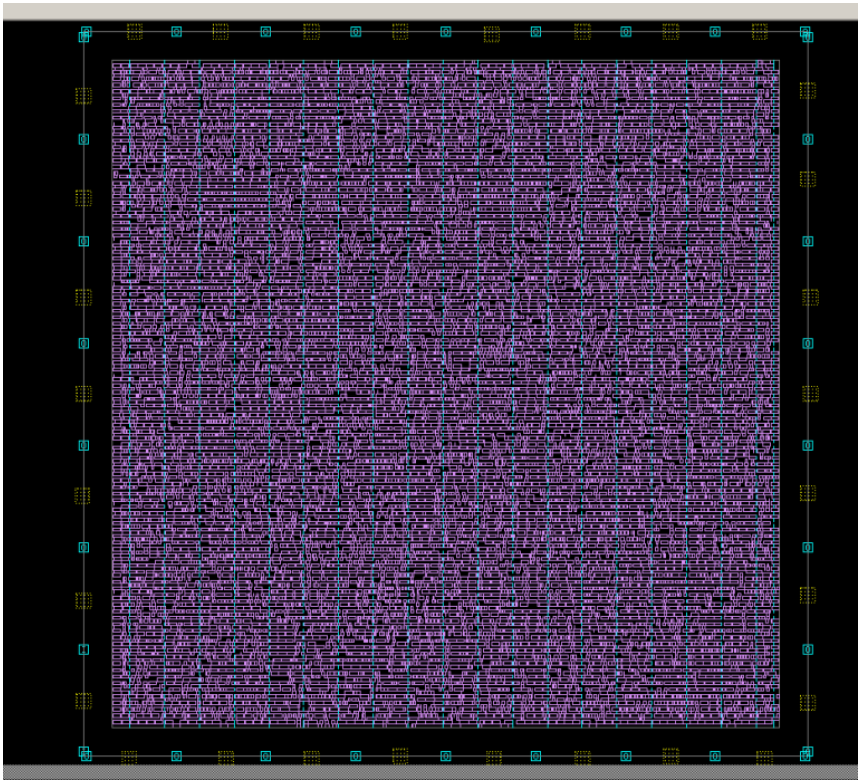


IR drop after power network

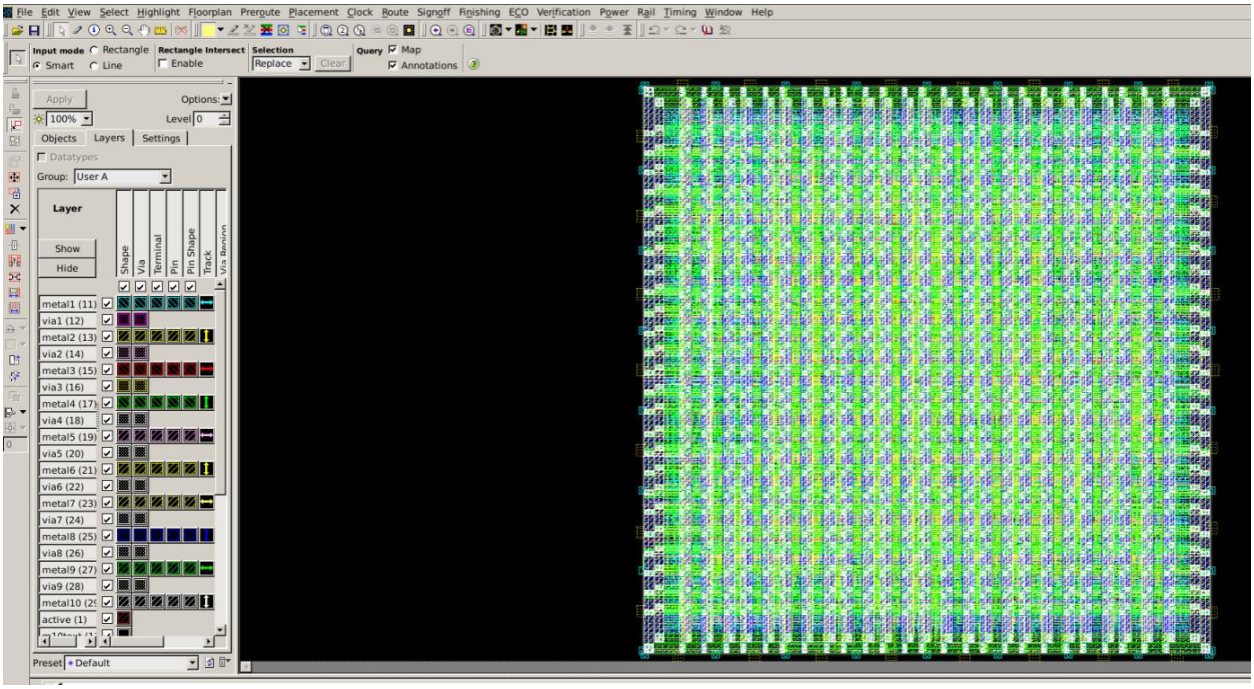


Before power network the IR drop in the center was higher “red”

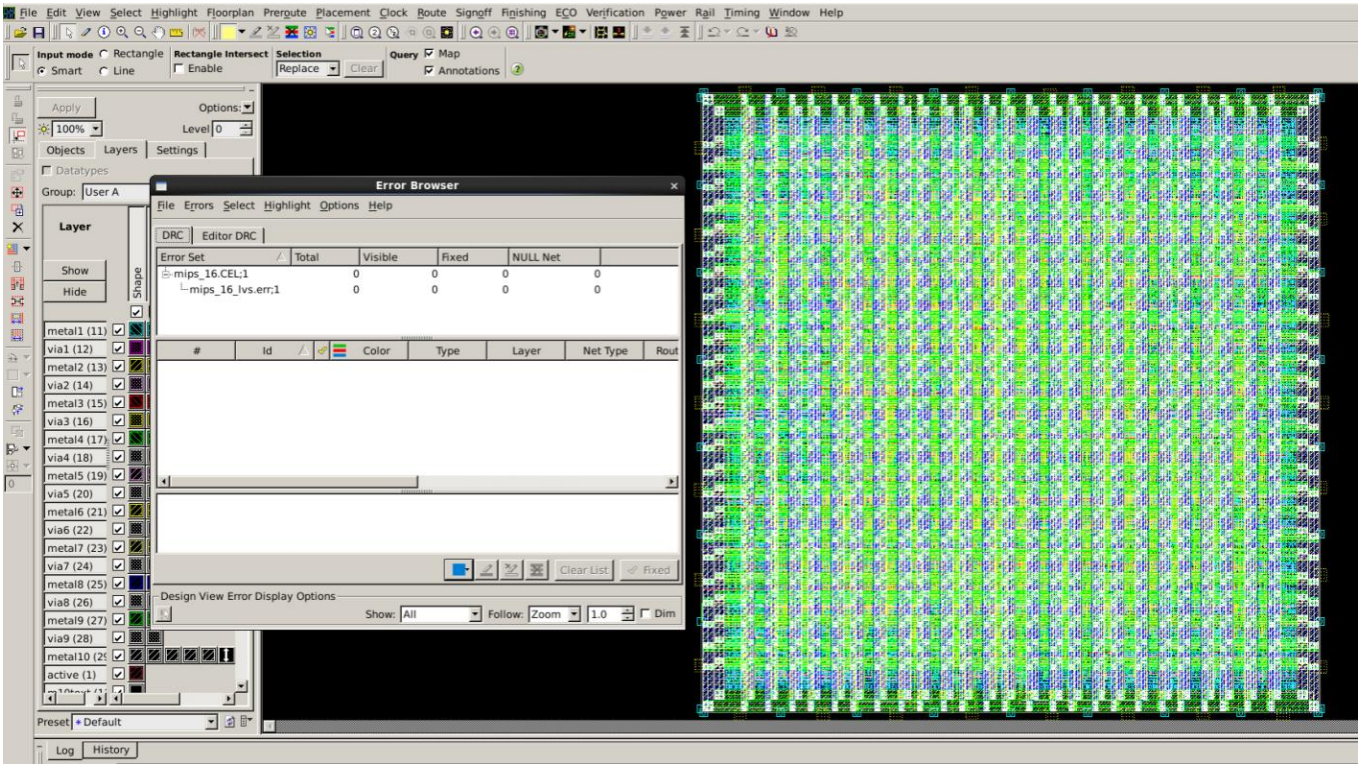
Placement of cell (after hiding the metal layers)



After cts and routing (hiding layers)



Final layout after chip finish (showing the layers)



Chapter 3 Conclusion

we met the specs of making the clock 400Mhz and the final area is less than 100,000 um^2, and there is no DRC violations and no setup and hold violations after the chip finish.

The final results folder (syn,formal,pnr) including inside them the reports and the scripts will be in the file in G02_Project2.zip folder.

Thank

You